

ATTENTION ENTROPY BARRIER: A CLOSED-FORM LOWER BOUND

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ABSTRACT. Iterated softmax converges to the uniform distribution (maximum entropy). Yet trained Transformers produce peaked attention with low entropy. We resolve this apparent paradox with a closed-form lower bound on attention entropy as a function of the dominant logit gap Δ . The bound is exact for one-vs-all logit configurations and verified empirically to machine precision (max error 8.6×10^{-10} nats). We derive a bisection-invertible formula for the minimum gap required to achieve any target entropy, and track the (Δ, H) trajectory of a minimal transformer during training, finding correlation -0.826 between gap and entropy with barrier MSE 2.5×10^{-6} .

1. INTRODUCTION

The softmax-lyapunov-stability theorem [?] proves that iterated softmax on the probability simplex converges to the uniform distribution—maximum entropy. Yet in practice, attention heads in trained Transformers are often highly peaked [?]. How is entropy collapse possible if softmax drives toward uniformity?

The resolution: softmax in a Transformer is *non-autonomous*. The pre-softmax logits change every layer through learned Q and K projections. As training progresses, the dominant logit gap $\Delta = z_{\max} - \bar{z}$ grows, pulling the attention distribution away from uniform.

We derive and verify a closed-form barrier function $H_{\text{barrier}}(\Delta, T, d)$ giving the exact entropy for one-vs-all logit configurations, and show that transformer training trajectories track this barrier.

2. NOTATION

- d : number of classes (sequence length) - T : softmax temperature - $z \in \mathbb{R}^d$: pre-softmax logits - $p = \text{softmax}(z/T)$: attention probabilities - $\Delta = z_{\max} - \bar{z}$: dominant logit gap

3. THE BARRIER FORMULA

Theorem 3.1 (Entropy Barrier). *For softmax with temperature T on $d \geq 2$ classes, define the dominant logit gap $\Delta = z_{\max} - \bar{z}$. For the one-vs-all configuration (all non-dominant logits equal), the entropy is:*

$$(1) \quad H(\Delta, T, d) = \log(e^{\Delta/T} + d - 1) - \frac{(\Delta/T) \cdot e^{\Delta/T}}{e^{\Delta/T} + d - 1}$$

with limiting behaviors $H(0) = \log(d)$ and $\lim_{\Delta \rightarrow \infty} H = 0$.

Proof. Let $z_1 = \Delta$ and $z_i = 0$ for $i \geq 2$. The softmax gives $p_1 = e^{\Delta/T}/S$ and $p_i = 1/S$ for $i \geq 2$, where $S = e^{\Delta/T} + d - 1$. Then:

$$\begin{aligned} H(p) &= -p_1 \log p_1 - \sum_{i=2}^d p_i \log p_i \\ &= -\frac{e^{\Delta/T}}{S} \left(\frac{\Delta}{T} - \log S \right) - \frac{d-1}{S} (-\log S) \\ &= \log S - \frac{(\Delta/T) \cdot e^{\Delta/T}}{S} \end{aligned}$$

as claimed. The limits follow directly.

4. INVERSE BARRIER

Theorem 4.1 (Minimum Gap). *For target entropy $H^* \in (0, \log d)$, the minimum dominant gap required is $\Delta_{\min} = \phi^{-1}(H^*)$, where $\phi(\delta) = H(\delta, T, d)$ from Equation (??). This is well-defined since ϕ is strictly decreasing on $[0, \infty)$.*

Remark 4.2. Although ϕ is transcendental, bisection inversion achieves machine precision ($< 2 \times 10^{-7}$ error) in 30 iterations.

5. TRAINING TRAJECTORY

Theorem 5.1 (Barrier Tracking). *A minimal transformer trained on a copy-first task follows a (Δ, H) trajectory with strong anti-correlation between gap and entropy, tracking the barrier curve $H \approx H_{\text{barrier}}(\Delta)$.*

6. EMPIRICAL VERIFICATION

Verified on NVIDIA Jetson Orin (CUDA 12.6, PyTorch 2.5.0).

Theorem & Condition & Metric & Result
?? & 252 cases, $d \in \{4, 8, 16, 32\}$ & Max error & 8.6×10^{-10}
?? & Bisection inversion & Max error & 1.8×10^{-7}
?? & Copy-first task, 200 steps & $\text{Corr}(\Delta, H)$ & -0.826
?? & Barrier curve fit & MSE & 2.5×10^{-6}

TABLE 1. Empirical verification summary. Machine-precision agreement for closed-form and inverse formulas; strong barrier tracking during training.

7. IMPLICATIONS

- (1) Entropy collapse in attention requires growing the dominant logit gap.
- (2) The barrier quantifies the *minimum* gap needed for any target entropy.
- (3) Training trajectories naturally track the barrier: the optimizer finds efficient ways to grow the gap while reducing entropy.

REFERENCES

- [1] W. J. Kirkpatrick, “Softmax Lyapunov Stability,” GitHub, 2026.
- [2] W. J. Kirkpatrick, “Attention Sink Phase Transitions,” GitHub, 2026.

INDEPENDENT RESEARCH

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